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ETSI

European Telecommunications Standards Institute

ETSI Secretariat

Postal address: F-06921 Sophia Antipolis CEDEX - FRANCE
Office address: 650 Route des Lucioles - Sophia Antipolis - Valbonne - FRANCE
X.400: c=fr, a=atlas, p=etsi, s=secretariat - **Internet:** secretariat@etsi.fr

Tel.: +33 92 94 42 00 - Fax: +33 93 65 47 16

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Foreword

This European Telecommunication Standard (ETS) has been produced by the Special Mobile Group (SMG) Technical Committee (TC) of the European Telecommunications Standards Institute (ETSI).

This ETS specifies the coding of information necessary for support of supplementary service operation on the mobile radio interface layer 3 within the European digital cellular telecommunications system (Phase 2).

This ETS correspond to GSM technical specification, GSM 04.80 version 4.9.2.

The specification from which this ETS has been derived was originally based on CEPT documentation, hence the presentation of this ETS may not be entirely in accordance with the ETSI/PNE rules.

Reference is made within this ETS to GSM Technical Specifications (GSM-TS) (NOTE).

Reference is also made within this ETS to GSM 04.8x series. The specifications in the series can be identified, with their full title, within the normative reference Clause of this ETS by the last two digits of their GSM reference number e.g. GSM 04.8x series, refers to GSM 04.81, GSM 04.82, etc.

NOTE: TC-SMG has produced documents which give the technical specifications for the implementation of the European digital cellular telecommunications system. Historically, these documents have been identified as GSM Technical Specifications (GSM-TS). These TSs may have subsequently become I-ETSs (Phase 1), or ETSs (Phase 2), whilst others may become ETSI Technical Reports (ETRs). GSM-TSs are, for editorial reasons, still referred to in current GSM ETSs.

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1 General

1.1 Scope

This technical specification contains the coding of information necessary for support of supplementary service operation on the mobile radio interface layer 3.

Section 2 gives the functional definitions and contents of messages for call independent supplementary service operations. Messages necessary for support of call related supplementary service operations are defined in TS GSM 04.08.

Section 3 gives the general format and coding for messages used for call independent supplementary service and the format and coding of information elements used for both call related and call independent supplementary service operations.

Section 4 gives the specification of the call related and call independent supplementary service operations.

1.2 Normative references

This ETS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

[1] GSM 01.04 (ETR 100): "European digital cellular telecommunications system (Phase 2); Abbreviations and acronyms".

[2] GSM 02.24 (ETS 300 510): "European digital cellular telecommunications system (Phase 2); Description of Charge Advice Information (CAI)".

[3] GSM 04.06 (ETS 300 555): "European digital cellular telecommunications system (Phase 2); Mobile Station - Base Station system (MS - BSS) interface Data Link (DL) layer specification".

[4] GSM 04.07 (ETS 300 556): "European digital cellular telecommunications system (Phase 2); Mobile radio interface signalling layer 3 General aspects".

[5] GSM 04.08 (ETS 300 557): "European digital cellular telecommunications system (Phase 2); Mobile radio interface layer 3 specification".

[6] GSM 04.10 (ETS 300 558): "European digital cellular telecommunications system (Phase 2); Mobile radio interface layer 3 Supplementary services specification General aspects".

[7] GSM 04.80 (ETS 300 564): "European digital cellular telecommunications system (Phase 2); Mobile radio interface layer 3 supplementary services specification Formats and coding".

[8] GSM 04.81 (ETS 300 565): "European digital cellular telecommunications system (Phase 2); Line identification supplementary services - Stage 3".

[9] GSM 04.82 (ETS 300 566): "European digital cellular telecommunications system (Phase 2); Call Forwarding (CF) supplementary services - Stage 3".

[10] GSM 04.83 (ETS 300 567): "European digital cellular telecommunications system (Phase 2); Call Waiting (CW) and Call Hold (HOLD) supplementary services - Stage 3".

[11] GSM 04.84 (ETS 300 568): "European digital cellular telecommunications system (Phase 2); Multi Party (MPTY) supplementary services - Stage 3".

[12] GSM 04.85 (ETS 300 569): "European digital cellular telecommunications system (Phase 2); Closed User Group (CUG) supplementary services - Stage 3".

[13] GSM 04.86 (ETS 300 570): "European digital cellular telecommunications system (Phase 2); Advice of Charge (AoC) supplementary services - Stage 3".

[14] GSM 04.88 (ETS 300 571): "European digital cellular telecommunications system (Phase 2); Call Barring (CB) supplementary services - Stage 3".

[15] GSM 04.90 (ETS 300 572): "European digital cellular telecommunications system (Phase 2); Unstructured supplementary services operation - Stage 3".

[16] GSM 09.02 (ETS 300 599): "European digital cellular telecommunications system (Phase 2); Mobile Application Part (MAP) specification".

[17] GSM 09.11 (ETS 300 606): "European digital cellular telecommunications system (Phase 2); Signalling interworking for supplementary services".

[18] CCITT Recommendation X.208: "Specification of basic encoding rules for Abstract Syntax Notation One (ASN.1)".

[19] CCITT Recommendation X.209: "Specification of Abstract Syntax Notation One (ASN.1)".

[20] CCITT Recommendation Q.773: "Transaction capabilities formats and encoding".

1.3 Definitions and abbreviations

Abbreviations used in this specification are listed in GSM 01.04.

2 Message functional definitions and contents

2.1 General

This section defines the structure of the messages of the layer 3 protocol defined in TS GSM 04.80. These messages are standard L3 messages as defined in TS GSM 04.07.

Each definition includes:

a) a brief description of the message;

b) a table listing the information elements in the order of their appearance in the message. In a sequence of consecutive IEs with half octet length, the first IE occupies bits 1 to 4 of octet N, the second bits 5 to 8 of octet N, the third bits 1 to 4 of octet N+1 etc;

   For each IE the table indicates:

   1) the information element identifier, in hexadecimal notation, if the IE has format T, TV or TLV. If the IEI has half octet length, it is specified by a notation representing the IEI as a hexadecimal digit followed by a "-" (example: B-);

   2) the name of the IE (which gives an idea of the semantics of the element), which is used in this and other specifications as a reference to the IE within the message;

   3) the name of the type of the IE (which indicates the coding of the value part of the IE), and a reference to a description of the value part of the IE;

   4) the presence requirement indication (M, C or O) for the IE, as defined in TS GSM 04.07;

   5) the format of the IE (T, V, TV, LV, TLV) as defined in TS GSM 04.07;

   6) the length of the IE (or permissible range of lengths), in octets, in the message, where "?" means that the maximum length of the IE is only constrained by the link layer protocol, and in the case of the facility IE by possible further considerations specified in TS GSM 04.10. This indication is non-normative.

c) subsections specifying conditions for IEs with presence requirement C or O in the relevant message. Together with other conditions specified in TS GSM 04.80, GSM 04.10 or GSM 04.8x and 04.9x-series this defines when the IE shall be included or not, what non-presence of such IEs means, and (for IEs with presence requirement C) the static conditions for presence and/or non-presence of the IEs (see TS GSM 04.07).

2.2 Messages for supplementary services control

Table 2.1 summarizes the messages for call independent supplementary services control. See TS GSM 04.10 for a detailed description of call independent supplementary service messages.

Messages for supplementary service control	Reference
FACILITY	2.3
REGISTER	2.4
RELEASE COMPLETE	2.5

Table 2.1: Messages for call independent supplementary service control

2.3 Facility

This message is sent by the mobile station or the network to request or acknowledge a supplementary service. It is used when information is to be conveyed and the transaction already exists, but is not to be released. The supplementary service to be invoked, and its associated parameters, are specified in the Facility information element. See table 2.2.

IEI	Information element	Type / Reference	Presence	Format	Length
	Supplementary service protocol discriminator	Protocol discriminator 3.2	M	V	1/2
	Transaction identifier	Transaction identifier 3.3	M	V	1/2
	Facility message type	Message type 3.4	M	V	1
	Facility	Facility 3.5	M	LV	2-?

Table 2.2: FACILITY message content

2.4 Register

2.4.1 Register (network to mobile station direction)

This message is sent by the network to the mobile station to assign a new transaction identifier for call independent supplementary service control and to request or acknowledge a supplementary service. See table 2.3.

IEI	Information element	Type / Reference	Presence	Format	Length
	Supplementary service protocol discriminator	Protocol discriminator 3.2	M	V	1/2
	Transaction identifier	Transaction identifier 3.3	M	V	1/2
	Register message type	Message type 3.4	M	V	1
1C	Facility	Facility 3.5	M	TLV	2-?

Table 2.3: REGISTER message content (network to mobile station direction)

2.4.2 Register (mobile station to network direction)

This message is sent by the mobile station to the network to assign a new transaction identifier for call independent supplementary service control and to request or acknowledge a supplementary service. See table 2.4.

IEI	Information element	Type / Reference	Presence	Format	Length
	Supplementary service protocol discriminator	Protocol discriminator 3.2	M	V	1/2
	Transaction identifier	Transaction identifier 3.3	M	V	1/2
	Register message type	Message type 3.4	M	V	1
1C	Facility	Facility 3.5	M	TLV	2-?
7F	SS version	SS version indicator 3.8.2	O	TLV	3

Table 2.4: REGISTER message content (mobile station to network direction)

2.4.2.1 SS version

This information element shall be included if the supplementary service operation being invoked is implemented according to the phase 2 GSM standards.

2.5 Release complete

This message is sent by the mobile station or the network to release a transaction used for call independent supplementary service control. It may also request or acknowledge a supplementary service. See table 2.5.

IEI	Information element	Type / Reference	Presence	Format	Length
	Supplementary service protocol discriminator	Protocol discriminator 3.2	M	V	1/2
	Transaction identifier	Transaction identifier 3.3	M	V	1/2
	Release Complete message type	Message type 3.4	M	V	1
08	Cause	Cause TS GSM 04.08	O	TLV	4-32
1C	Facility	Facility 3.5	O	TLV	2-?

Table 2.5: RELEASE COMPLETE message content

2.5.1 Cause

This information element shall be included when the functional handling of the Cause IE is specified in the service description or TS GSM 09.11. If the functional handling of the Cause IE is not specified, the receiving entity may ignore the IE.

2.5.2 Facility

This information element shall be included as required by the service description and the procedures defined in TS GSM 04.10.

3 General message format and information elements coding

The figures and text in this section describe message contents. Within each octet, the bit designated "bit 1" is transmitted first, followed by bits 2, 3, 4, etc. Similarly, the octet shown at the top of each figure is sent first.

3.1 Overview

Within the layer 3 protocol defined in TS GSM 04.80, every message is a standard L3 message as defined in TS GSM 04.07. This means that the message consists of the following parts:

a) protocol discriminator;
b) transaction identifier;
c) message type;
d) other information elements, as required.

Unless specified otherwise, a particular information element may be present only once in a given message.

When a field extends over more than one octet, the order of bit values progressively decreases as the octet number increases. The least significant bit of the field is represented by the lowest numbered bit of the highest numbered octet of the field.

3.2 Protocol discriminator

The Protocol Discriminator (PD) and its use are defined in TS GSM 04.07. TS GSM 04.80 defines the protocols relating to the PD values:

1 0 1 1 supplementary services (call independent).

3.3 Transaction identifier

For general rules, format and coding of transaction identifier values, see TS GSM 04.08.

3.4 Message type

The message type IE and its use are defined in TS GSM 04.07. Table 3.1 defines the value part of the message type IE used in the supplementary service protocol.

8 7 6 5 4 3 2 1	**Message types**
0 x 1 0	Clearing messages:
1 0 1 0	- RELEASE COMPLETE
0 x 1 1	Miscellaneous message group:
1 0 1 0	- FACILITY
1 0 1 1	- REGISTER

NOTE 1: Bit 8 is reserved for possible future use as an extension bit, see TS GSM 04.07.

NOTE 2: Bit 7 is reserved for the send sequence number in messages sent from the mobile station. In messages sent from the network, bit 7 is coded with a "0", see TS GSM 04.07.

Table 3.1: Message types

3.5 Other information elements

These information elements are coded according to the general coding rules as defined in TS GSM 04.08.

Table 3.2 contains the code-points allocated to the information elements used in messages defined in this technical specification. All IEs are defined in TS GSM 04.08, but the content of the Facility and SS version indicator IEs are defined within this specification.

8 7 6 5 4 3 2 1		**Reference (IE content)**
0	Type 3 and 4 information elements	
0 0 0 1 0 0 0	Cause	TS GSM 04.08
0 0 1 1 1 0 0	Facility	3.6
1 1 1 1 1 1 1	SS version indicator	3.8.2

Table 3.2: Information elements specific to call independent supplementary service control

3.6 Facility information element

The purpose of the Facility information element is to indicate the invocation and operation of supplementary services, identified by the corresponding operation code within the Facility information element.

The Facility information element is coded as shown in figure 3.1 and tables 3.3 to 3.17.

The Facility is a type 4 information element with no upper length limit except that given by the maximum number of octets in a L3 message, see TS GSM 04.06.

8	**7**	**6**	**5**	**4**	**3**	**2**	**1**	
0	0	0	1	1	1	0	0	octet 1
Facility IEI								
Length of Facility contents								octet 2
Component(s) (NOTE)								octet 3 etc.

NOTE: One or more components may be included depending on specific service requirements.

Figure 3.1: Facility information element

3.6.1 Component (octet 3 etc)

This section provides the formats and encoding of components in the Facility information element. Formats and encoding methods make use of and is a subset of CCITT Recommendation Q.773 (Transaction Capabilities formats and Encoding) and T/S 43/BB. The used part of CCITT Recommendation Q.773 respectively T/S 43/BB is almost the same as the Component Portion of TC messages. The only difference is that returnResultNotLast is not used.

This section is further based on:

- CCITT Recommendation X.208 (Specification of Abstract Syntax Notation One (ASN.1));

- CCITT Recommendation X.209 (Specification of basic encoding rules for Abstract Syntax Notation One);

and is consistent with these CCITT recommendations.

The CCITT Recommendations X.208 and X.209 formal description language is not used.

The parameters in tables 3.3 to 3.6 may be one of the following:

- a Sequence of Parameters;

- a Set of Parameters;

- a specific Parameter with its own tag (i.e. not part of a Sequence or Set);

- nothing at all (i.e. absent).

NOTE: Concerning the general rules for encoding (structure of encoding, identifier octets, length octets, etc.) see CCITT Recommendations X.208 and X.209. For these general rules the same exceptions apply as stated in TS GSM 09.02. This holds also for tables 3.3 to 3.6.

Invoke component	**Reference**	**Mandatory indication**
Component type tag Component length	3.6.2 X.209	M
Invoke ID tag Invoke ID length Invoke ID	3.6.3 X.209 3.6.3	M
Linked ID tag Linked ID length Linked ID	3.6.3 X.209 3.6.3	O
Operation Code tag Operation Code length Operation Code	3.6.4 X.209 3.6.4	M
Parameters	4	O

Table 3.3: Invoke component

Return Result component	**Reference**	**Mandatory indication**
Component type tag Component length	3.6.2 X.209	M
Invoke ID tag Invoke ID length Invoke ID	3.6.3 X.209 3.6.3	M
Sequence tag Sequence length	3.6.5 X.209	O (NOTE)
Operation Code tag Operation Code length Operation Code	3.6.4 X.209 3.6.4	O (NOTE)
Parameters	4	O (NOTE)

NOTE: Omitted if the Return Result component does not include any parameters.

Table 3.4: Return Result component

Return Error component	**Reference**	**Mandatory indication**
Component type tag Component length	3.6.2 X.209	M
Invoke ID tag Invoke ID length Invoke ID	3.6.3 X.209 3.6.3	M
Error Code tag Error Code length Error Code	3.6.6 X.209 3.6.6	M
Parameters	4	O

Table 3.5: Return Error component

Reject component	Reference	Mandatory indication
Component type tag Component length	3.6.2 X.209	M
Invoke ID tag (NOTE) Invoke ID length Invoke ID	3.6.3 X.209 3.6.3	M
Problem Code tag Problem Code length Problem Code	3.6.7 X.209 3.6.7	M

NOTE: If the Invoke ID is not available, Universal Null (table 3.9) with length = 0 shall be used.

Table 3.6: Reject component

3.6.2 Component type tag

The Component type tag is coded context-specific, constructor as indicated in table 3.7.

Component type tag	**8 7 6 5 4 3 2 1**
Invoke	1 0 1 0 0 0 0 1
Return Result	1 0 1 0 0 0 1 0
Return Error	1 0 1 0 0 0 1 1
Reject	1 0 1 0 0 1 0 0

Table 3.7: Coding of Component type tag

3.6.3 Component ID tag

The term Component ID refers to the Invoke ID or the Linked ID. The Component ID tag is coded as shown in table 3.8.

Component ID tag	**8 7 6 5 4 3 2 1**
Invoke ID	0 0 0 0 0 0 1 0
Linked ID (NOTE)	1 0 0 0 0 0 0 0

NOTE: This tag differs from the Invoke ID tag, which is coded as a Universal INTEGER, in order to distinguish it from the following tag (Operation Code) which is also coded as a Universal INTEGER.

Table 3.8: Coding of Component ID tag

The length of a Component ID is 1 octet.

An Invoke Component has one or two Component IDs: an Invoke ID, and if it is desired to associate the Invoke with a previous Invoke, then the Linked ID is provided in addition to the Invoke ID.

Return Result and Return Error Components have one Component ID, called an Invoke ID which is the reflection of the Invoke ID of the Invoke Component to which they are responding.

The Reject Component uses as its Invoke ID, the Invoke ID in the Component being rejected. If this ID is unavailable (e.g. due to mutilation of the message not detected by lower layers), then the Invoke ID tag is replaced with a universal NULL tag as shown in table 3.9. Universal NULL has always length = 0

Any kind of component, except a reject component, may be rejected.

	8 7 6 5 4 3 2 1
NULL tag	0 0 0 0 0 1 0 1

Table 3.9: Coding of NULL tag

If an Invoke containing both Invoke and Linked IDs is being rejected, only the Invoke ID is used in the Reject Component.

3.6.4 Operation Code

Each Operation is assigned an Operation Code to identify it. An Operation Code follows an Operation Code tag and Operation Code length. The Operation Code tag is coded as shown in table 3.10.

	8 7 6 5 4 3 2 1
Operation Code tag	0 0 0 0 0 0 1 0

Table 3.10: Coding of Operation Code tag

The Operation Codes for the different Operations are defined in section 4.5.

3.6.5 Sequence and Set tags

When there is more than one parameter in a Component (applicable to all Component types), they follow the Sequence or Set tag, which are coded universal, constructor as shown in table 3.11.

Sequence and set tags	**8 7 6 5 4 3 2 1**
Sequence tag	0 0 1 1 0 0 0 0
Set tag	0 0 1 1 0 0 0 1

Table 3.11: Coding of Sequence and set tags

3.6.6 Error Code

Each Error is assigned a value (Error Code) to identify it.

An Error Code follows an Error Code tag and Error Code length. The Error Code tag is coded as shown in table 3.12.

	8 7 6 5 4 3 2 1
Error Code tag	0 0 0 0 0 0 1 0

Table 3.12: Coding of Error Code tag

The Error Codes for the different Errors are defined in section 4.5.

3.6.7 Problem Code

The Problem Code consists of one of the four elements: General Problem, Invoke Problem, Return Result Problem or Return Error Problem. The tags for these elements are coded as shown in table 3.13.

Problem tags	**8 7 6 5 4 3 2 1**
General Problem tag	1 0 0 0 0 0 0 0
Invoke Problem tag	1 0 0 0 0 0 0 1
Return Result Problem tag	1 0 0 0 0 0 1 0
Return Error Problem tag	1 0 0 0 0 0 1 1

Table 3.13: Coding of Problem tags

The Problem Codes for the different Problems are shown in tables 3.14 to 3.17.

General Problem Codes	**8 7 6 5 4 3 2 1**
Unrecognized Component	0 0 0 0 0 0 0 0
Mistyped Component	0 0 0 0 0 0 0 1
Badly Structured Component	0 0 0 0 0 0 1 0

Table 3.14: Coding of General Problem Codes

Invoke Problem Codes	**8 7 6 5 4 3 2 1**
Duplicate Invoke ID	0 0 0 0 0 0 0 0
Unrecognized Operation	0 0 0 0 0 0 0 1
Mistyped Parameter	0 0 0 0 0 0 1 0
Resource Limitation	0 0 0 0 0 0 1 1
Initiating Release	0 0 0 0 0 1 0 0
Unrecognized Linked ID	0 0 0 0 0 1 0 1
Linked Response Unexpected	0 0 0 0 0 1 1 0
Unexpected Linked Operation	0 0 0 0 0 1 1 1

Table 3.15: Coding of Invoke Problem Codes

Return Result Problem Codes	**8 7 6 5 4 3 2 1**
Unrecognized Invoke ID	0 0 0 0 0 0 0 0
Return Result Unexpected	0 0 0 0 0 0 0 1
Mistyped Parameter	0 0 0 0 0 0 1 0

Table 3.16: Coding of Return Result Problem Codes

Return Error Problem Codes	**8 7 6 5 4 3 2 1**
Unrecognized Invoke ID	0 0 0 0 0 0 0 0
Return Error Unexpected	0 0 0 0 0 0 0 1
Unrecognized Error	0 0 0 0 0 0 1 0
Unexpected Error	0 0 0 0 0 0 1 1
Mistyped Parameter	0 0 0 0 0 1 0 0

Table 3.17: Coding of Return Error Problem Codes

3.7 Version handling for supplementary services

3.7.1 Supplementary service screening indicator

The purpose of the supplementary service screening indicator is to allow the network to asses the capabilities of the MS in advance of a network initiated SS activity. The SS screening indicator is sent in the mobile station classmark 2 as defined in TS GSM 04.08. The handling of the SS screening indicator is described in TS GSM 04.10.

8	7	6	5	4	3	2	1
(NOTE)	SS screening indicator			(NOTE)			

NOTE: Values not relevant to supplementary services.

Figure 3.2: Coding of SS screening indicator in mobile station classmark 2

SS screening indicator in mobile station classmark 2	6	5
default value of phase 1	0	0
capability of handling of ellipsis notation and phase 2 error handling (NOTE 1)	0	1
for future use (NOTE 2)	1	0
for future use (NOTE 2)	1	1

NOTE 1: Ellipsis notation is described in TS GSM 04.10 and GSM 09.02. SS Error handling is described in TS GSM 04.10.

NOTE 2: The network shall interpret these values the same as "01".

Table 3.18: Coding of SS screening indicator in mobile station classmark 2

3.7.2 Supplementary service version indicator

The purpose of the supplementary service version indicator is to allow the network to select the correct version of a protocol for a specific supplementary service. The SS version indicator is included in messages as defined in TS GSM 04.08 and GSM 04.80. The coding described in table 3.19 refers to the first octet received in the SS version indicator. Any other octets received shall be ignored. The handling of the SS version indicator is described in TS GSM 04.10.

SS version indicator	8	7	6	5	4	3	2	1
phase 2 service, ellipsis notation, and phase 2 error handling is supported (NOTE 1)	0	0	0	0	0	0	0	0
all other values are for future use (NOTE 2)								

NOTE 1: Ellipsis notation is described in TS GSM 04.10 and GSM 09.02. SS Error handling is described in TS GSM 04.10.

NOTE 2: The network shall interpret all possible values of the SS version indicator the same as "00000000".

Table 3.19: Coding of SS version indicator

4 Supplementary services operation specifications

4.1 General

This section specifies the abstract syntax for the Supplementary Service protocol using the Abstract Syntax Notation One (ASN.1), defined in CCITT Recommendation X.208 (1998).

The mapping of OPERATION and ERROR to components is defined in section 3 of this technical specification.

The encoding rules which are applicable to the defined abstract syntax are the Basic Encoding Rules for Abstract Syntax Notation One, defined in CCITT Recommendation X.209 (1998) with the same exceptions as stated in TS GSM 09.02. For each Supplementary Service parameter which has to be transferred by a Supplementary Service message, there is a PDU field (an ASN.1 NamedType) whose ASN.1 identifier has the same name as the corresponding parameter, except for the differences required by the ASN.1 notation (blanks between words are removed, the first letter of the first word is lower-case and the first letter of the following words are capitalized (e.g "bearer service" is mapped to "bearerService"). In addition some words may be abbreviated as follows:

- ms mobile subscriber;
- ss supplementary services;
- cug closed user group.

The ASN.1 data type which follows the keywords ARGUMENT "PARAMETER" or "RESULT" (for OPERATION and ERROR) is always optional from a syntactic point of view. However, except specific mention, it has to be considered as mandatory from a semantic point of view. When in an invoke component, a mandatory element is missing in any component or inner data structure, a reject component is returned with the problem code "Mistyped Parameter". When an optional element is missing in an invoke component or in an inner data structure while it is required by the context, an error component is returned; the associated type of error is "DataMissing".

In case an element is defined as mandatory in the protocol description (TS GSM 04.80 including imports from TS GSM 09.02), but is not present according to the service description (stage 1 to stage 3), the ASN.1 protocol description takes precedence over the diagrams in the GSM 04.8x and 04.9x-series of technical specifications.

When possible operations and errors are imported from TS GSM 09.02 thereby making the MSC transparent to most of the messages sent to or from the MS.

Timer values for operations which require timers are shown as ASN.1 comments.

Ellipsis Notation shall be used in the same way as described in TS GSM 09.02 and shall be supported on the radio interface by the MS and the network for all operations defined in this specification including those imported from TS GSM 09.02.

4.2 Operation types

Table 4.1 summarises the operations defined for supplementary services in this technical specification and shows which of these operations are call related and call independent. The terms "call related" and "call independent" are defined in TS GSM 04.10.

Operation name	Call related SS	Call independent SS
RegisterSS	-	+
EraseSS	-	+
ActivateSS	-	+
DeactivateSS	-	+
InterrogateSS	-	+
RegisterPassword	-	+
GetPassword	-	+
ProcessUnstructuredSS-Data	+	+
ForwardCheckSS-Indication	-	+
ProcessUnstructuredSS-Request	-	+
UnstructuredSS-Request	-	+
UnstructuredSS-Notify	-	+
ForwardChargeAdvice	+	-
NotifySS	+	-
ForwardCUG-Info	+	-
BuildMPTY	+	-
HoldMPTY	+	-
RetrieveMPTY	+	-
SplitMPTY	+	-

NOTE: The ProcessUnstructuredSS-Data operation may be used call related by a phase 1 MS.

Table 4.1: Relevance of supplementary service operations

```
SS-Operations {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-Operations (0) version2 (2)}

DEFINITIONS ::=

BEGIN

EXPORTS

-- exports operation types

-- operations defined in this technical specification
NotifySS, ForwardChargeAdvice, ForwardCUG-Info, BuildMPTY, HoldMPTY, RetrieveMPTY, SplitMPTY;

IMPORTS

OPERATION FROM
TCAPMessages {
  ccitt recommendation q 773 modules (2) messages (1) version2 (2)}
```

```
-- The MAP operations
-- RegisterSS, EraseSS, ActivateSS, DeactivateSS, InterrogateSS,
-- RegisterPassword, GetPassword, ProcessUnstructuredSS-Data,
-- ProcessUnstructuredSS-Request, UnstructuredSS-Request, UnstructuredSS-Notify
-- ForwardCheckSS-Indication
-- are imported from MAP-Operations in SS-Protocol module.

-- imports SS-data types
NotifySS-Arg,
ForwardChargeAdviceArg,
ForwardCUG-InfoArg
FROM SS-DataTypes {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-Datatypes (2) version2 (2)}

-- imports MAP-errors
IllegalSS-Operation, SS-ErrorStatus, SS-NotAvailable,
SS-Incompatibility, SystemFailure, FacilityNotSupported
FROM MAP-Errors {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
  map-Errors (10) version2 (2)}

-- imports SS-Errors
ResourcesNotAvailable, MaxNumberOfMPTY-ParticipantsExceeded
FROM SS-Errors {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-Errors (1) version2 (2)}
;

-- operation types definition

NotifySS ::=                    OPERATION
      ARGUMENT
      notifySS-Arg              NotifySS-Arg

ForwardChargeAdvice ::=         OPERATION -- Timer T(AoC)= 1s to 40s
      ARGUMENT
      forwardChargeAdviceArg    ForwardChargeAdviceArg
      RESULT

ForwardCUG-Info ::=             OPERATION
      ARGUMENT
      forwardCUG-InfoArg        ForwardCUG-InfoArg

BuildMPTY ::=                   OPERATION -- Timer T(BuildMPTY)= 5s to 30s
      RESULT
      ERRORS{
        IllegalSS-Operation,
        SS-ErrorStatus,
        SS-NotAvailable,
        SS-Incompatibility,
        SystemFailure,
        ResourcesNotAvailable,
        MaxNumberOfMPTY-ParticipantsExceeded}
```

```
HoldMPTY ::=                    OPERATION -- Timer T(HoldMPTY)= 5s to 30s
      RESULT
      ERRORS{
        IllegalSS-Operation,
        SS-ErrorStatus,
        SS-Incompatibility,
        FacilityNotSupported,
        SystemFailure}

RetrieveMPTY ::=                OPERATION -- Timer T(RetrieveMPTY)= 5s to 30s
      RESULT
      ERRORS{
        IllegalSS-Operation,
        SS-ErrorStatus,
        SS-Incompatibility,
        FacilityNotSupported,
        SystemFailure}

SplitMPTY ::=                   OPERATION -- Timer T(SplitMPTY)= 5s to 30s
      RESULT
      ERRORS{
        IllegalSS-Operation,
        SS-ErrorStatus,
        SS-Incompatibility,
        FacilityNotSupported,
        SystemFailure}

END
```

4.2.1 [spare]

4.2.2 Operation types description

For each operation type this section provides a brief prose description.

4.2.2.1 RegisterSS (MS --> network)

This operation type is invoked by an MS to register data related to a supplementary service in the network. When no BasicService parameter is provided, the registration applies to all provisioned and applicable basic services.

4.2.2.2 EraseSS (MS --> network)

This operation type is invoked by an MS to erase data related to a supplementary service in the network. When no BasicService parameter is provided, the erasure applies to all provisioned and applicable basic services.

4.2.2.3 ActivateSS (MS --> network)

This operation type is invoked by an MS to request the network for a supplementary service activation. When no BasicService parameter is provided, the activation applies to all provisioned and applicable basic services.

4.2.2.4 DeactivateSS (MS --> network)

This operation type is invoked by an MS to request the network for a supplementary service deactivation. When no BasicService parameter is provided, the deactivation applies to all provisioned and applicable basic services.

4.2.2.5 InterrogateSS (MS --> network)

This operation type is invoked by an MS to request the network for a supplementary service interrogation. When no BasicService parameter is provided, the interrogation applies to all provisioned and applicable basic services.

4.2.2.6 NotifySS (network --> MS)

This operation type is invoked by the network to forward a supplementary service notification towards a mobile subscriber.

4.2.2.7 RegisterPassword (MS --> network)

This operation type is invoked by an MS to register a new password related to the management by the subscriber himself of subscription data in the HLR. The operation "Register password" will be successful if the subscriber can provide the old password, the new password and the new password again as results of 3 subsequent operations "Get password".

4.2.2.8 GetPassword (network --> MS)

This operation type is invoked by the network to request a password from the mobile subscriber. It may be used to allow the registration of a new password or the management of subscription data by the subscriber himself (e.g. modification of call barring activation status).

4.2.2.9 ProcessUnstructuredSS-Data (MS --> network)

This operation type is invoked by an MS to relay unstructured information in order to allow end to end SS operation between the MS and the HLR following specific rules (e.g. embedding of keypad commands). The operation is used in order to provide backward compatibility (see TS GSM 04.90).

4.2.2.10 ProcessUnstructuredSS-Request (MS --> network)

This operation type is invoked by an MS to start an unstructured supplementary service data application in the network.

4.2.2.11 UnstructuredSS-Request (network --> MS)

This operation type is invoked by the network to request unstructured information from the MS in order to perform an unstructured supplementary service data application.

4.2.2.12 UnstructuredSS-Notify (network --> MS)

This operation type is invoked by the network to give an unstructured supplementary service notification to the mobile user.

4.2.2.13 ForwardCheckSSIndication (network --> MS)

This operation type is invoked by the network to indicate to the mobile subscriber that the status of supplementary services may not be correct in the network. The procedures for initiating ForwardCheckSSIndication are specified in TS GSM 09.02.

4.2.2.14 ForwardChargeAdvice (network --> MS)

This operation type is invoked by the network to forward Advice of Charge information to the mobile subscriber.

4.2.2.15 BuildMPTY (MS --> network)

This operation type is invoked by an MS to request the network to connect calls in a multi party call.

4.2.2.16 HoldMPTY (MS --> network)

This operation type is invoked by an MS to put the MS-connection to a multi party call (invoked by that MS) on hold.

4.2.2.17 RetrieveMPTY (MS --> network)

This operation type is invoked by an MS to request retrieval of a multi party call held by that MS.

4.2.2.18 SplitMPTY (MS --> network)

This operation type is invoked by an MS to request a private communication with one of the remote parties in a multi party call invoked by that MS.

4.2.2.19 ForwardCUG-Info (MS --> network)

This operation type is used by an MS to explicitly invoke a CUG call.

4.3 Error types

4.3.1 Error types ASN.1 specification

The following ASN.1 module provides an ASN.1 specification of errors. Errors from MAP are imported in the SS-Protocol module in section 4.5.

```
SS-Errors {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-Errors (1) version2 (2)}

DEFINITIONS ::=

BEGIN

IMPORTS

ERROR FROM
TCAPMessages {
  ccitt recommendation q 773 modules (2) messages (1) version2 (2)};

-- The MAP errors
-- UnknownSubscriber, BearerServiceNotProvisioned, TeleserviceNotProvisioned,
-- IllegalSS-Operation, SS-ErrorStatus, SS-NotAvailable, SS-SubscriptionViolation,
-- SS-Incompatibility, SystemFailure, DataMissing, UnexpectedDataValue, FacilityNotSupported,
-- PW-RegistrationFailure, NegativePW-Check, CallBarred, NumberOfPW-AttemptsViolation,
-- AbsentSubscriber, IllegalSubscriber, IllegalEquipment, USSD-Busy, UnknownAlphabet
-- are imported from MAP-Errors in SS-Protocol module.

-- error types definition
ResourcesNotAvailable ::= ERROR
MaxNumberOfMPTY-ParticipantsExceeded ::= ERROR

END
```

4.3.2 Error types description

For each error type this section provides a brief prose description.

4.3.2.1 UnknownSubscriber

This error is returned by the network when it is requested to perform an operation concerning an unknown subscriber.

4.3.2.2 BearerServiceNotProvisioned

This error is returned by the network when it is requested to perform an operation on a supplementary service and not even a subset of the requested bearer service group has been subscribed to.

4.3.2.3 TeleServiceNotProvisioned

This error is returned by the network when it is requested to perform an operation on a supplementary service and not even a subset of the requested teleservice group has been subscribed to.

4.3.2.4 IllegalSS-Operation

This error is returned by the network when it is requested to perform an illegal operation which is defined as not applicable for the relevant supplementary service(s) (e.g. registration request for a service which must be registered by the administration). For the definition of the allowed operations for the individual supplementary services, see GSM 04.8x and 04.9x-series of technical specifications.

4.3.2.5 SS-ErrorStatus

This error is returned by the network when it is requested to perform an operation which is not compatible with the current status of the relevant supplementary service. The current status may be given as additional information by use of the SS-parameter.

4.3.2.6 SS-NotAvailable

This error is returned by the network when it is requested to perform an operation on a supplementary service which is not available in the current location area.

4.3.2.7 SS-SubscriptionViolation

This error is returned by the network when it is requested to perform an operation on a supplementary service, transgressing the subscription restrictions. The nature of the restriction or the transgressed options may be sent as parameters.

4.3.2.8 SS-Incompatibility

This error is returned by the network when it is requested for a supplementary service operation incompatible with the status of an other supplementary service or with the teleservice or bearer service for which the operation is requested. This error shall only be used if the operation is not compatible for even a subset of the teleservice group or bearer service group specified in the request. The identity and status of the conflicting service may also be indicated. The additional information may contain the SS-code parameter, the Basic Service Group parameter and the SS-status parameter.

4.3.2.9 SystemFailure

This error is returned by the network, when it cannot perform an operation because of a failure in the network.

4.3.2.10 DataMissing

This error is returned by the network when an optional parameter is missing in an invoke component or an inner data structure, while it is required by the context of the request.

4.3.2.11 UnexpectedDataValue

This error is returned by the network when it receives a parameter with an unexpected value, without type violation.

4.3.2.12 PasswordRegistrationFailure

This error is returned when a password registration procedure fails because of abnormal subscriber inputs. A more specific diagnostic may be passed as error parameter and indicates situations such as:

- invalid password format;
- new passwords mismatch.

4.3.2.13 NegativePasswordCheck

This error is returned to indicate the negative result of a password check because the subscriber has not provided the required password or has provided a password which does not match the valid one.

4.3.2.14 FacilityNotSupported

This error is returned by the network receiving a request about a facility which is not supported in the PLMN.

4.3.2.15 ResourcesNotAvailable

This error is returned by the network to the MS if temporarily there are no resources to support e.g. a multi party call available in the network.

4.3.2.16 MaxNumberOfMPTY-ParticipantsExceeded

This error is returned by the network to the MS if the request must be rejected because the number of subscribers to join a multi party call would exceed the maximum value.

4.3.2.17 CallBarred

This error is returned by the network to the MS when call independent subscriber control procedures are barred by the operator. The parameter "operator barring" shall be included.

4.3.2.18 NumberOfPW-AttemptsViolation

This error is returned by the network to the MS when the maximum number of wrong password attempts is exceeded.

4.3.2.19 AbsentSubscriber

This error is returned when the subscriber has activated the detach service or the system detects the absence condition. This error is not used on the radio interface but only between network entities.

4.3.2.20 IllegalSubscriber

This error is returned when illegality of the access has been established by use of authentication procedure. This error is not used on the radio interface but only between network entities.

4.3.2.21 IllegalEquipment

This error is returned when the IMEI check procedure has shown that the IMEI is blacklisted or not white-listed. This error is not used on the radio interface but only between network entities.

4.3.2.22 USSD-Busy

This error is returned by the MS to the network when the MS is not able to process the unstructured supplementary service data operation due to an on-going MMI input of the user or an already existing call independent supplementary service transaction.

4.3.2.23 UnknownAlphabet

This error is returned by the MS or the network when the alphabet/language used for the unstructured supplementary service data operation is not known by the network or the MS.

4.4 Data types and identifiers

4.4.1 General

The data types used in the SS protocol specifications are described in the ASN.1 module provided in section 4.4.2, while section 4.4.3 provides an overview of the identifiers used in SS ASN.1 specifications.

Since size constraints are subject to modifications named values have been defined in the following module for the upper boundaries of the value ranges associated to several sub-type specifications.

4.4.2 ASN.1 data types

This section provides an ASN.1 module defining the abstract data types in operations and errors specification. Only data types which are specific for this technical specification are defined. All other data types are imported from MAP together with the import of operations and errors.

```
SS-DataTypes {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-DataTypes (2) version2 (2)}

DEFINITIONS

IMPLICIT TAGS ::=

BEGIN

-- exports all data types defined in this module

IMPORTS

SS-Code
FROM MAP-SS-Code {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
  map-SS-Code (15) version2 (2)}

-- imports MAP-SS-DataTypes
SS-Status, CUG-Index
FROM MAP-SS-DataTypes {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
  map-SS-DataTypes (14) version2 (2)}
;

-- data types definition

NotifySS-Arg ::= SEQUENCE{
      ss-Code                       [1]     SS-Code OPTIONAL,
      ss-Status                     [4]     SS-Status OPTIONAL,
      ss-Notification               [5]     SS-Notification OPTIONAL,
      callIsWaiting-Indicator       [14]    NULL OPTIONAL,
      callOnHold-Indicator          [15]    CallOnHold-Indicator OPTIONAL,
      mpty-Indicator                [16]    NULL OPTIONAL,
      cug-Index                     [17]    CUG-Index OPTIONAL,
      clirSuppressionRejected       [18]    NULL OPTIONAL,
      ...}
```

```
ForwardChargeAdviceArg ::= SEQUENCE{
    ss-Code                         [0]     SS-Code,
    chargingInformation             [1]     ChargingInformation,
    ...}

SS-Notification ::= OCTET STRING (SIZE (1))

--  Bit 8 7 6 5 4           00000 (Unused)

--      Bit 3               Call is forwarded indication to A-subscriber
--                          (calling subscriber)
--          0               No information content
--          1               Outgoing call has been forwarded to C

--      Bit 2               Call is forwarded indication to B-subscriber
--                          (forwarding subscriber)
--          0               No information content
--          1               Incoming call has been forwarded to C

--      Bit 1               Call is forwarded indication to C-subscriber
--                          (forwarded-to subscriber)
--          0               No information content
--          1               Incoming call is a forwarded call

ChargingInformation ::= SEQUENCE{
    e1                              [1] E1 OPTIONAL,
    e2                              [2] E2 OPTIONAL,
    e3                              [3] E3 OPTIONAL,
    e4                              [4] E4 OPTIONAL,
    e5                              [5] E5 OPTIONAL,
    e6                              [6] E6 OPTIONAL,
    e7                              [7] E7 OPTIONAL,
    ...}

E1 ::= INTEGER (0..max10TimesUnitsPerTime)
max10TimesUnitsPerTime INTEGER ::= 8191

E2 ::= INTEGER (0..max10TimesTimeInterval)
max10TimesTimeInterval INTEGER ::= 8191

E3 ::= INTEGER (0..max100TimesScalingFactor)
max100TimesScalingFactor INTEGER ::= 8191

E4 ::= INTEGER (0..max10TimesIncrement)
max10TimesIncrement INTEGER ::= 8191

E5 ::= INTEGER (0..max10TimesIncrementPerDataInterval)
max10TimesIncrementPerDataInterval INTEGER ::= 8191

E6 ::= INTEGER (0..maxNumberOfSegmentsPerDataInterval)
maxNumberOfSegmentsPerDataInterval INTEGER ::= 8191

E7 ::= INTEGER (0..max10TimesInitialTime)
max10TimesInitialTime INTEGER ::= 8191
```

```
CallOnHold-Indicator            ::= ENUMERATED {
                                        callRetrieved (0),
                                        callOnHold (1)}

ForwardCUG-InfoArg ::= SEQUENCE {
     cug-Index                          [0]     CUG-Index OPTIONAL,
     suppressPrefCUG                    [1]     NULL OPTIONAL,
     suppressOA                         [2]     NULL OPTIONAL,
     ...}

END
```

4.4.3 Identifiers definition

The parameters which are described in the following subsections correspond to the identifiers used in operation and error types description.

4.4.3.1 chargingInformation

The chargingInformation identifier refers to the necessary information for the Advice of Charge supplementary service. See TS GSM 02.24.

4.4.3.2 e1

The e1 identifier refers to 10 times the number of LPLMN units per time interval in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.3 e2

The e2 identifier refers to 10 times the length of the time interval in seconds in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.4 e3

The e3 identifier refers to 100 times the scaling factor to convert from LPLMN units to HPLMN units in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.5 e4

The e4 identifier refers to 10 times the LPLMN increment in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.6 e5

The e5 identifier refers to 10 times the number of LPLMN units incremented per data interval in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.7 e6

The e6 identifier refers to the number of segments per data interval in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.8 e7

The e7 identifier refers to 10 times the length of the initial time interval in seconds in connection with the Advice of Charge supplementary service, see TS GSM 02.24.

4.4.3.9 ss-Code

The ss-Code identifier refers to the code which identify a supplementary service or a group of supplementary services.

4.4.3.10 ss-Notification

The ss-Notification identifier refers to one or several supplementary service notifications which have to be forwarded to a mobile subscriber.

4.4.3.11 ss-Status

The ss-Status identifier refers to the status of a supplementary service.

4.4.3.12 callIsWaiting-Indicator

The callIsWaiting-Indicator identifier refers to the indication given to the mobile station that the call is waiting.

4.4.3.13 callOnhold-Indicator

The callOnHold-Indicator identifier refers to the indication given to the mobile station that the call has been put on hold or has been retrieved.

4.4.3.14 mpty-Indicator

The mpty-Indicator identifier refers to the indication given to the mobile station that the multi party call has been invoked.

4.4.3.15 forwardCUG-InfoArg

The forwardCUG-InfoArg identifier refers to the indication given from the mobile subscriber to the network in connection with explicit invocation of a CUG call.

4.4.3.16 cug-Index

The cug-Index identifier refers to the index of a CUG given in an explicit invocation of a CUG call.

4.4.3.17 suppressPrefCUG

The suppressPrefCUG identifier refers to the mobile subscribers request to the network to prohibit the use of the preferential CUG.

4.4.3.18 suppressOA

The suppressOA identifier refers to the mobile subscribers request to the network to prohibit the use of the subscriber option "OA allowed".

4.4.3.19 clirSuppressionRejected

The clirSuppressionRejected identifier refers to the indication given to the mobile station that the CLIR suppression request has been rejected.

4.5 Operations and errors implementation

For the actual implementation of supplementary services, operations and errors have to be defined by value. The following ASN.1 module, imports operation types from the ASN.1 module described in section 4.2 and operation and error types from MAP. It defines operations by allocating operations and errors a local value. For the involved operations and errors the same local values as in MAP are allocated.

```
SS-Protocol {
  ccitt identified-organization (4) etsi (0) mobileDomain (0)
  gsm-Access (2) modules (3) ss-Protocol (3) version2 (2)}

DEFINITIONS ::=

BEGIN

IMPORTS

-- imports operation types

-- imports operation type from MAP-MobileServiceOperations
ForwardCheckSS-Indication
FROM MAP-MobileServiceOperations {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
  map-MobileServiceOperations (5) version2 (2)}

-- imports operation types from MAP-SupplementaryServiceOperations
RegisterSS, EraseSS, ActivateSS, DeactivateSS, InterrogateSS, RegisterPassword, GetPassword,
ProcessUnstructuredSS-Data, ProcessUnstructuredSS-Request, UnstructuredSS-Request,
UnstructuredSS-Notify
FROM MAP-SupplementaryServiceOperations {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
  map-SupplementaryServiceOperations (8) version2 (2)}

-- imports operation types from SS-Operations
NotifySS, ForwardChargeAdvice, BuildMPTY, HoldMPTY, RetrieveMPTY, SplitMPTY,
ForwardCUG-Info
FROM SS-Operations {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-Operations (0) version2 (2)}

-- imports error types

-- imports error types from MAP-Errors
UnknownSubscriber, BearerServiceNotProvisioned, TeleserviceNotProvisioned,
IllegalSS-Operation, SS-ErrorStatus, SS-NotAvailable, SS-SubscriptionViolation,
SS-Incompatibility, SystemFailure, DataMissing, UnexpectedDataValue, PW-RegistrationFailure,
NegativePW-Check, FacilityNotSupported, CallBarred, NumberOfPW-AttemptsViolation,
AbsentSubscriber, IllegalSubscriber, IllegalEquipment, USSD-Busy, UnknownAlphabet


FROM MAP-Errors {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
  map-Errors (10) version2 (2)}
```

```
-- imports error types from SS-Errors
ResourcesNotAvailable, MaxNumberOfMPTY-ParticipantsExceeded
FROM SS-Errors {
  ccitt identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3)
  ss-Errors (1) version2 (2)}
;

-- allocation of local values to operations

registerSS                          RegisterSS ::= localValue 10
eraseSS                             EraseSS ::= localValue 11
activateSS                          ActivateSS ::= localValue 12
deactivateSS                        DeactivateSS ::= localValue 13
interrogateSS                       InterrogateSS ::= localValue 14
notifySS                            NotifySS ::= localValue 16
registerPassword                    RegisterPassword ::= localValue 17
getPassword                         GetPassword ::= localValue 18
processUnstructuredSS-Data          ProcessUnstructuredSS-Data ::= localValue 19
forwardCheckSS-Indication           ForwardCheckSS-Indication ::= localValue 38
processUnstructuredSS-Request       ProcessUnstructuredSS-Request ::= localValue 59
unstructuredSS-Request              UnstructuredSS-Request ::= localValue 60
unstructuredSS-Notify               UnstructuredSS-Notify ::= localValue 61
forwardCUG-Info                     ForwardCUG-Info ::= localValue 120
splitMPTY                           SplitMPTY ::= localValue 121
retrieveMPTY                        RetrieveMPTY ::= localValue 122
holdMPTY                            HoldMPTY ::= localValue 123
buildMPTY                           BuildMPTY ::= localValue 124
forwardChargeAdvice                 ForwardChargeAdvice ::= localValue 125

-- allocation of local values to errors

unknownSubscriber                   UnknownSubscriber ::= localValue 1
illegalSubscriber                   IllegalSubscriber ::= localValue 9
bearerServiceNotProvisioned         BearerServiceNotProvisioned ::= localValue 10
teleserviceNotProvisioned           TeleserviceNotProvisioned ::= localValue 11
illegalEquipment                    IllegalEquipment ::= localValue 12
callBarred                          CallBarred ::= localValue 13
illegalSS-Operation                 IllegalSS-Operation ::= localValue 16
ss-ErrorStatus                      SS-ErrorStatus ::= localValue 17
ss-NotAvailable                     SS-NotAvailable ::= localValue 18
ss-SubscriptionViolation            SS-SubscriptionViolation ::= localValue 19
ss-Incompatibility                  SS-Incompatibility ::= localValue 20
facilityNotSupported                FacilityNotSupported ::= localValue 21
absentSubscriber                    AbsentSubscriber ::= localValue 27
systemFailure                       SystemFailure ::= localValue 34
dataMissing                         DataMissing ::= localValue 35
unexpectedDataValue                 UnexpectedDataValue ::= localValue 36
pw-RegistrationFailure              PW-RegistrationFailure ::= localValue 37
negativePW-Check                    NegativePW-Check ::= localValue 38
numberOfPW-AttemptsViolation        NumberOfPW-AttemptsViolation ::= localValue 43
unknownAlphabet                     UnknownAlphabet ::= localValue 71
ussd-Busy                           USSD-Busy ::= localValue 72
maxNumberOfMPTY-ParticipantsExceeded
                                    MaxNumberOfMPTY-ParticipantsExceeded ::= localValue 126
resourcesNotAvailable               ResourcesNotAvailable ::= localValue 127

END
```

Annex A (informative): Expanded ASN.1 Module "SS-Protocol"

```
--          Expanded ASN.1 Module 'SS-Protocol'
--SIEMENS ASN.1 Compiler       P2.10  (94-10-21 10:03:57)
--            Date: 94-10-21 Time: 10:50:58

SS-Protocol { 0 identified-organization (4) etsi (0) mobileDomain (0) gsm-Access (2) modules (3) ss-
Protocol (3) version2 (2) }

DEFINITIONS

::=

BEGIN

registerSS OPERATION
  ARGUMENT
    registerSS-Arg SEQUENCE {
      ss-Code    OCTET STRING (SIZE (1)),
      basicService CHOICE {
        bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
        teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
      forwardedToNumber [4] IMPLICIT OCTET STRING (SIZE (1..20)) OPTIONAL,
      forwardedToSubaddress [6] IMPLICIT OCTET STRING (SIZE (1..21)) OPTIONAL,
      noReplyConditionTime [5] IMPLICIT INTEGER (5..30) OPTIONAL,
      ... }
  RESULT
    ss-Info    CHOICE {
      forwardingInfo [0] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        forwardingFeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            forwardedToNumber [5] IMPLICIT OCTET STRING (SIZE (1..9)) OPTIONAL,
            forwardedToSubaddress [8] IMPLICIT OCTET STRING (SIZE (1..21)) OPTIONAL,
            forwardingOptions [6] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            noReplyConditionTime [7] IMPLICIT INTEGER (5..30) OPTIONAL,
            ... },
        ... },
      callBarringInfo [1] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        callBarringFeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            ... },
        ... },
      cug-Info   [2] IMPLICIT SEQUENCE {
        cug-SubscriptionList SEQUENCE SIZE (1..10) OF
          SEQUENCE {
            cug-Index  INTEGER (0..32767),
            cug-Interlock OCTET STRING (SIZE (4)),
            intraCUG-Options ENUMERATED {
              noCUG-Restrictions  (0),
              cugIC-CallBarred  (1),
              cugOG-CallBarred  (2)},
            basicServiceGroupList SEQUENCE SIZE (1..13) OF
              CHOICE {
```

```
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              ... },
          cug-FeatureList SEQUENCE SIZE (1..13) OF
            SEQUENCE {
              basicService CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              preferentialCUG-Indicator INTEGER (0..32767) OPTIONAL,
              interCUG-Restrictions OCTET STRING (SIZE (1)),
              ... } OPTIONAL,
          ... },
        ss-Data    [3] IMPLICIT SEQUENCE {
          ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
          ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
          ss-SubscriptionOption CHOICE {
            cliRestrictionOption [2] IMPLICIT ENUMERATED {
              permanent   (0),
              temporaryDefaultRestricted  (1),
              temporaryDefaultAllowed  (2)},
            overrideCategory [1] IMPLICIT ENUMERATED {
              overrideEnabled  (0),
              overrideDisabled  (1)}} OPTIONAL,
          basicServiceGroupList SEQUENCE SIZE (1..13) OF
            CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
          ... }}
    ERRORS {
      -- systemFailure -- localValue 34,
      -- dataMissing -- localValue 35,
      -- unexpectedDataValue -- localValue 36,
      -- unknownSubscriber -- localValue 1,
      -- bearerServiceNotProvisioned -- localValue 10,
      -- teleserviceNotProvisioned -- localValue 11,
      -- callBarred -- localValue 13,
      -- illegalSS-Operation -- localValue 16,
      -- ss-ErrorStatus -- localValue 17,
      -- ss-SubscriptionViolation -- localValue 19,
      -- ss-Incompatibility -- localValue 20}
   ::= localValue 10

eraseSS    OPERATION
    ARGUMENT
      ss-ForBS   SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)),
        basicService CHOICE {
          bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
          teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
        ... }
    RESULT
      ss-Info    CHOICE {
        forwardingInfo [0] IMPLICIT SEQUENCE {
          ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
          forwardingFeatureList SEQUENCE SIZE (1..13) OF
            SEQUENCE {
              basicService CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
              forwardedToNumber [5] IMPLICIT OCTET STRING (SIZE (1..9)) OPTIONAL,
              forwardedToSubaddress [8] IMPLICIT OCTET STRING (SIZE (1..21)) OPTIONAL,
              forwardingOptions [6] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
```

```
            noReplyConditionTime [7] IMPLICIT INTEGER (5..30) OPTIONAL,
            ... },
        ... },
      callBarringInfo [1] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        callBarringFeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            ... },
        ... },
      cug-Info   [2] IMPLICIT SEQUENCE {
        cug-SubscriptionList SEQUENCE SIZE (1..10) OF
          SEQUENCE {
            cug-Index  INTEGER (0..32767),
            cug-Interlock OCTET STRING (SIZE (4)),
            intraCUG-Options ENUMERATED {
              noCUG-Restrictions  (0),
              cugIC-CallBarred  (1),
              cugOG-CallBarred  (2)},
            basicServiceGroupList SEQUENCE SIZE (1..13) OF
              CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ... },
        cug-FeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            preferentialCUG-Indicator INTEGER (0..32767) OPTIONAL,
            interCUG-Restrictions OCTET STRING (SIZE (1)),
            ... } OPTIONAL,
        ... },
      ss-Data    [3] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
        ss-SubscriptionOption CHOICE {
          cliRestrictionOption [2] IMPLICIT ENUMERATED {
            permanent   (0),
            temporaryDefaultRestricted  (1),
            temporaryDefaultAllowed  (2)},
          overrideCategory [1] IMPLICIT ENUMERATED {
            overrideEnabled  (0),
            overrideDisabled  (1)}} OPTIONAL,
        basicServiceGroupList SEQUENCE SIZE (1..13) OF
          CHOICE {
            bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
            teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
        ... }}
  ERRORS {
    -- systemFailure -- localValue 34,
    -- dataMissing -- localValue 35,
    -- unexpectedDataValue -- localValue 36,
    -- unknownSubscriber -- localValue 1,
    -- bearerServiceNotProvisioned -- localValue 10,
    -- teleserviceNotProvisioned -- localValue 11,
    -- callBarred -- localValue 13,
    -- illegalSS-Operation -- localValue 16,
    -- ss-ErrorStatus -- localValue 17,
    -- ss-SubscriptionViolation -- localValue 19}
```

```
 ::= localValue 11

activateSS OPERATION
  ARGUMENT
    ss-ForBS   SEQUENCE {
      ss-Code    OCTET STRING (SIZE (1)),
      basicService CHOICE {
        bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
        teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
      ... }
  RESULT
    ss-Info    CHOICE {
      forwardingInfo [0] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        forwardingFeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            forwardedToNumber [5] IMPLICIT OCTET STRING (SIZE (1..9)) OPTIONAL,
            forwardedToSubaddress [8] IMPLICIT OCTET STRING (SIZE (1..21)) OPTIONAL,
            forwardingOptions [6] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            noReplyConditionTime [7] IMPLICIT INTEGER (5..30) OPTIONAL,
            ... },
        ... },
      callBarringInfo [1] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        callBarringFeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            ... },
        ... },
      cug-Info   [2] IMPLICIT SEQUENCE {
        cug-SubscriptionList SEQUENCE SIZE (1..10) OF
          SEQUENCE {
            cug-Index  INTEGER (0..32767),
            cug-Interlock OCTET STRING (SIZE (4)),
            intraCUG-Options ENUMERATED {
              noCUG-Restrictions  (0),
              cugIC-CallBarred  (1),
              cugOG-CallBarred  (2)},
            basicServiceGroupList SEQUENCE SIZE (1..13) OF
              CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ... },
        cug-FeatureList SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            preferentialCUG-Indicator INTEGER (0..32767) OPTIONAL,
            interCUG-Restrictions OCTET STRING (SIZE (1)),
            ... } OPTIONAL,
        ... },
      ss-Data    [3] IMPLICIT SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
        ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
        ss-SubscriptionOption CHOICE {
```

```
            cliRestrictionOption [2] IMPLICIT ENUMERATED {
              permanent   (0),
              temporaryDefaultRestricted  (1),
              temporaryDefaultAllowed  (2)},
            overrideCategory [1] IMPLICIT ENUMERATED {
              overrideEnabled  (0),
              overrideDisabled  (1)}} OPTIONAL,
          basicServiceGroupList SEQUENCE SIZE (1..13) OF
            CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
          ... }}
    ERRORS {
      -- systemFailure -- localValue 34,
      -- dataMissing -- localValue 35,
      -- unexpectedDataValue -- localValue 36,
      -- unknownSubscriber -- localValue 1,
      -- bearerServiceNotProvisioned -- localValue 10,
      -- teleserviceNotProvisioned -- localValue 11,
      -- callBarred -- localValue 13,
      -- illegalSS-Operation -- localValue 16,
      -- ss-ErrorStatus -- localValue 17,
      -- ss-SubscriptionViolation -- localValue 19,
      -- ss-Incompatibility -- localValue 20,
      -- negativePW-Check -- localValue 38,
      -- numberOfPW-AttemptsViolation -- localValue 43}
  ::= localValue 12
deactivateSS OPERATION
    ARGUMENT
      ss-ForBS   SEQUENCE {
        ss-Code    OCTET STRING (SIZE (1)),
        basicService CHOICE {
          bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
          teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
        ... }
    RESULT
      ss-Info    CHOICE {
        forwardingInfo [0] IMPLICIT SEQUENCE {
          ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
          forwardingFeatureList SEQUENCE SIZE (1..13) OF
            SEQUENCE {
              basicService CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
              forwardedToNumber [5] IMPLICIT OCTET STRING (SIZE (1..9)) OPTIONAL,
              forwardedToSubaddress [8] IMPLICIT OCTET STRING (SIZE (1..21)) OPTIONAL,
              forwardingOptions [6] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
              noReplyConditionTime [7] IMPLICIT INTEGER (5..30) OPTIONAL,
              ... },
          ... },
        callBarringInfo [1] IMPLICIT SEQUENCE {
          ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
          callBarringFeatureList SEQUENCE SIZE (1..13) OF
            SEQUENCE {
              basicService CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
              ... },
          ... },
        cug-Info   [2] IMPLICIT SEQUENCE {
          cug-SubscriptionList SEQUENCE SIZE (1..10) OF
```

```
            SEQUENCE {
              cug-Index  INTEGER (0..32767),
              cug-Interlock OCTET STRING (SIZE (4)),
              intraCUG-Options ENUMERATED {
                noCUG-Restrictions  (0),
                cugIC-CallBarred  (1),
                cugOG-CallBarred  (2)},
              basicServiceGroupList SEQUENCE SIZE (1..13) OF
                CHOICE {
                  bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                  teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              ... },
          cug-FeatureList SEQUENCE SIZE (1..13) OF
            SEQUENCE {
              basicService CHOICE {
                bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
                teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
              preferentialCUG-Indicator INTEGER (0..32767) OPTIONAL,
              interCUG-Restrictions OCTET STRING (SIZE (1)),
              ... } OPTIONAL,
          ... },
        ss-Data    [3] IMPLICIT SEQUENCE {
          ss-Code    OCTET STRING (SIZE (1)) OPTIONAL,
          ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
          ss-SubscriptionOption CHOICE {
            cliRestrictionOption [2] IMPLICIT ENUMERATED {
              permanent   (0),
              temporaryDefaultRestricted  (1),
              temporaryDefaultAllowed  (2)},
            overrideCategory [1] IMPLICIT ENUMERATED {
              overrideEnabled  (0),
              overrideDisabled  (1)}} OPTIONAL,
          basicServiceGroupList SEQUENCE SIZE (1..13) OF
            CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
          ... }}
  ERRORS {
    -- systemFailure -- localValue 34,
    -- dataMissing -- localValue 35,
    -- unexpectedDataValue -- localValue 36,
    -- unknownSubscriber -- localValue 1,
    -- bearerServiceNotProvisioned -- localValue 10,
    -- teleserviceNotProvisioned -- localValue 11,
    -- callBarred -- localValue 13,
    -- illegalSS-Operation -- localValue 16,
    -- ss-ErrorStatus -- localValue 17,
    -- ss-SubscriptionViolation -- localValue 19,
    -- negativePW-Check -- localValue 38,
    -- numberOfPW-AttemptsViolation -- localValue 43}
 ::= localValue 13

interrogateSS OPERATION
  ARGUMENT
    ss-ForBS   SEQUENCE {
      ss-Code    OCTET STRING (SIZE (1)),
      basicService CHOICE {
        bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
        teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
      ... }
  RESULT
    interrogateSS-Res CHOICE {
      ss-Status  [0] IMPLICIT OCTET STRING (SIZE (1)),
```

```
        forwardedToNumber [1] IMPLICIT OCTET STRING (SIZE (1..9)),
        basicServiceGroupList [2] IMPLICIT SEQUENCE SIZE (1..13) OF
          CHOICE {
            bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
            teleservice [3] IMPLICIT OCTET STRING (SIZE (1))},
        forwardingFeatureList [3] IMPLICIT SEQUENCE SIZE (1..13) OF
          SEQUENCE {
            basicService CHOICE {
              bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
              teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
            ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            forwardedToNumber [5] IMPLICIT OCTET STRING (SIZE (1..9)) OPTIONAL,
            forwardedToSubaddress [8] IMPLICIT OCTET STRING (SIZE (1..21)) OPTIONAL,
            forwardingOptions [6] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
            noReplyConditionTime [7] IMPLICIT INTEGER (5..30) OPTIONAL,
            ... },
        cli-RestrictionInfo [4] IMPLICIT SEQUENCE {
          ss-Status  OCTET STRING (SIZE (1)),
          cliRestrictionOption ENUMERATED {
            permanent   (0),
            temporaryDefaultRestricted  (1),
            temporaryDefaultAllowed  (2)} OPTIONAL,
          ... }}
  ERRORS {
    -- systemFailure -- localValue 34,
    -- dataMissing -- localValue 35,
    -- unexpectedDataValue -- localValue 36,
    -- unknownSubscriber -- localValue 1,
    -- bearerServiceNotProvisioned -- localValue 10,
    -- teleserviceNotProvisioned -- localValue 11,
    -- callBarred -- localValue 13,
    -- illegalSS-Operation -- localValue 16,
    -- ss-NotAvailable -- localValue 18}
 ::= localValue 14

notifySS   OPERATION
  ARGUMENT
    notifySS-Arg SEQUENCE {
      ss-Code    [1] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
      ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
      ss-Notification [5] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
      callIsWaiting-Indicator [14] IMPLICIT NULL OPTIONAL,
      callOnHold-Indicator [15] IMPLICIT ENUMERATED {
        callRetrieved  (0),
        callOnHold  (1)} OPTIONAL,
      mpty-Indicator [16] IMPLICIT NULL OPTIONAL,
      cug-Index  [17] IMPLICIT INTEGER (0..32767) OPTIONAL,
      clirSuppressionRejected [18] IMPLICIT NULL OPTIONAL,
      ... }
 ::= localValue 16

registerPassword OPERATION
  ARGUMENT
    ss-Code    OCTET STRING (SIZE (1))
  RESULT
    newPassword NumericString  (FROM ("0"|"1"|"2"|"3"|"4"|"5"|"6"|"7"|"8"|"9")|SIZE (4))
  ERRORS {
    -- systemFailure -- localValue 34,
    -- dataMissing -- localValue 35,
    -- unexpectedDataValue -- localValue 36,
    -- callBarred -- localValue 13,
    -- ss-SubscriptionViolation -- localValue 19,
    -- pw-RegistrationFailure -- localValue 37,
```

```
      -- negativePW-Check -- localValue 38,
      -- numberOfPW-AttemptsViolation -- localValue 43}
    LINKED {
      -- getPassword -- localValue 18}
 ::= localValue 17

getPassword OPERATION
    ARGUMENT
      guidanceInfo ENUMERATED {
        enterPW     (0),
        enterNewPW  (1),
        enterNewPW-Again  (2),
        badPW-TryAgain  (3),
        badPW-FormatTryAgain  (4)}
    RESULT
   currentPassword NumericString  (FROM ("0"|"1"|"2"|"3"|"4"|"5"|"6"|"7"|"8"|"9")|SIZE (4))
 ::= localValue 18

processUnstructuredSS-Data OPERATION
    ARGUMENT
      ss-UserData IA5String   (SIZE (1..200))
    RESULT
      ss-UserData IA5String   (SIZE (1..200))
    ERRORS {
      -- systemFailure -- localValue 34,
      -- unexpectedDataValue -- localValue 36}
 ::= localValue 19

forwardCheckSS-Indication OPERATION
 ::= localValue 38

processUnstructuredSS-Request OPERATION
    ARGUMENT
      ussd-Arg   SEQUENCE {
        ussd-DataCodingScheme OCTET STRING (SIZE (1)),
        ussd-String OCTET STRING (SIZE (1..160)),
        ... }
    RESULT
      ussd-Res   SEQUENCE {
        ussd-DataCodingScheme OCTET STRING (SIZE (1)),
        ussd-String OCTET STRING (SIZE (1..160)),
        ... }
    ERRORS {
      -- systemFailure -- localValue 34,
      -- dataMissing -- localValue 35,
      -- unexpectedDataValue -- localValue 36,
      -- unknownAlphabet -- localValue 71,
      -- callBarred -- localValue 13}
 ::= localValue 59

unstructuredSS-Request OPERATION
    ARGUMENT
      ussd-Arg   SEQUENCE {
        ussd-DataCodingScheme OCTET STRING (SIZE (1)),
        ussd-String OCTET STRING (SIZE (1..160)),
        ... }
    RESULT
      ussd-Res   SEQUENCE {
        ussd-DataCodingScheme OCTET STRING (SIZE (1)),
        ussd-String OCTET STRING (SIZE (1..160)),
        ... }
    ERRORS {
      -- systemFailure -- localValue 34,
```

```
      -- dataMissing -- localValue 35,
      -- unexpectedDataValue -- localValue 36,
      -- absentSubscriber -- localValue 27,
      -- illegalSubscriber -- localValue 9,
      -- illegalEquipment -- localValue 12,
      -- unknownAlphabet -- localValue 71,
      -- ussd-Busy -- localValue 72}
 ::= localValue 60

unstructuredSS-Notify OPERATION
   ARGUMENT
      ussd-Arg   SEQUENCE {
         ussd-DataCodingScheme OCTET STRING (SIZE (1)),
         ussd-String OCTET STRING (SIZE (1..160)),
         ... }
   ERRORS {
      -- systemFailure -- localValue 34,
      -- dataMissing -- localValue 35,
      -- unexpectedDataValue -- localValue 36,
      -- absentSubscriber -- localValue 27,
      -- illegalSubscriber -- localValue 9,
      -- illegalEquipment -- localValue 12,
      -- unknownAlphabet -- localValue 71,
      -- ussd-Busy -- localValue 72}
 ::= localValue 61

forwardCUG-Info OPERATION
   ARGUMENT
      forwardCUG-InfoArg SEQUENCE {
         cug-Index  [0] IMPLICIT INTEGER (0..32767) OPTIONAL,
         suppressPrefCUG [1] IMPLICIT NULL OPTIONAL,
         suppressOA [2] IMPLICIT NULL OPTIONAL,
         ... }
 ::= localValue 120
splitMPTY  OPERATION
   RESULT
      zzzz-empty NULL
   ERRORS {
      -- illegalSS-Operation -- localValue 16,
      -- ss-ErrorStatus -- localValue 17,
      -- ss-Incompatibility -- localValue 20,
      -- facilityNotSupported -- localValue 21,
      -- systemFailure -- localValue 34}
 ::= localValue 121

retrieveMPTY OPERATION
   RESULT
      zzzz-empty NULL
   ERRORS {
      -- illegalSS-Operation -- localValue 16,
      -- ss-ErrorStatus -- localValue 17,
      -- ss-Incompatibility -- localValue 20,
      -- facilityNotSupported -- localValue 21,
      -- systemFailure -- localValue 34}
 ::= localValue 122
holdMPTY   OPERATION
   RESULT
      zzzz-empty NULL
   ERRORS {
      -- illegalSS-Operation -- localValue 16,
      -- ss-ErrorStatus -- localValue 17,
      -- ss-Incompatibility -- localValue 20,
      -- facilityNotSupported -- localValue 21,
```

```
      -- systemFailure -- localValue 34}
 ::= localValue 123

buildMPTY  OPERATION
   RESULT
      zzzz-empty NULL
   ERRORS {
      -- illegalSS-Operation -- localValue 16,
      -- ss-ErrorStatus -- localValue 17,
      -- ss-NotAvailable -- localValue 18,
      -- ss-Incompatibility -- localValue 20,
      -- systemFailure -- localValue 34,
      -- resourcesNotAvailable -- localValue 127,
      -- maxNumberOfMPTY-ParticipantsExceeded -- localValue 126}
 ::= localValue 124

forwardChargeAdvice OPERATION
   ARGUMENT
      forwardChargeAdviceArg SEQUENCE {
         ss-Code    [0] IMPLICIT OCTET STRING (SIZE (1)),
         chargingInformation [1] IMPLICIT SEQUENCE {
            e1       [1] IMPLICIT INTEGER (0..8191) OPTIONAL,
            e2       [2] IMPLICIT INTEGER (0..8191) OPTIONAL,
            e3       [3] IMPLICIT INTEGER (0..8191) OPTIONAL,
            e4       [4] IMPLICIT INTEGER (0..8191) OPTIONAL,
            e5       [5] IMPLICIT INTEGER (0..8191) OPTIONAL,
            e6       [6] IMPLICIT INTEGER (0..8191) OPTIONAL,
            e7       [7] IMPLICIT INTEGER (0..8191) OPTIONAL,
            ... },
         ... }
   RESULT
      zzzz-empty NULL
 ::= localValue 125

unknownSubscriber ERROR
 ::= localValue 1

illegalSubscriber ERROR
 ::= localValue 9

bearerServiceNotProvisioned ERROR
 ::= localValue 10

teleserviceNotProvisioned ERROR
 ::= localValue 11

illegalEquipment ERROR
 ::= localValue 12

callBarred ERROR
   PARAMETER
      callBarringCause ENUMERATED {
         barringServiceActive  (0),
         operatorBarring  (1)}
 ::= localValue 13

illegalSS-Operation ERROR
 ::= localValue 16
ss-ErrorStatus ERROR
   PARAMETER
      ss-Status  OCTET STRING (SIZE (1))
 ::= localValue 17
```

```
ss-NotAvailable ERROR
 ::= localValue 18

ss-SubscriptionViolation ERROR
   PARAMETER
      ss-SubscriptionOption CHOICE {
         cliRestrictionOption [2] IMPLICIT ENUMERATED {
            permanent   (0),
            temporaryDefaultRestricted  (1),
            temporaryDefaultAllowed  (2)},
         overrideCategory [1] IMPLICIT ENUMERATED {
            overrideEnabled  (0),
            overrideDisabled  (1)}}
 ::= localValue 19

ss-Incompatibility ERROR
   PARAMETER
      ss-IncompatibilityCause SEQUENCE {
         ss-Code    [1] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
         basicService CHOICE {
            bearerService [2] IMPLICIT OCTET STRING (SIZE (1)),
            teleservice [3] IMPLICIT OCTET STRING (SIZE (1))} OPTIONAL,
         ss-Status  [4] IMPLICIT OCTET STRING (SIZE (1)) OPTIONAL,
         ... }
 ::= localValue 20

facilityNotSupported ERROR
 ::= localValue 21

absentSubscriber ERROR
   PARAMETER
      mwd-Set    BOOLEAN
 ::= localValue 27
systemFailure ERROR
   PARAMETER
      networkResource ENUMERATED {
         plmn       (0),
         hlr        (1),
         vlr        (2),
         pvlr       (3),
         controllingMSC  (4),
         vmsc       (5),
         eir        (6),
         rss        (7)}
 ::= localValue 34

dataMissing ERROR
 ::= localValue 35

unexpectedDataValue ERROR
 ::= localValue 36

pw-RegistrationFailure ERROR
   PARAMETER
      pw-RegistrationFailureCause ENUMERATED {
         undetermined  (0),
         invalidFormat  (1),
         newPasswordsMismatch  (2)}
 ::= localValue 37

negativePW-Check ERROR
 ::= localValue 38
```

```
numberOfPW-AttemptsViolation ERROR
 ::= localValue 43

unknownAlphabet ERROR
 ::= localValue 71

ussd-Busy  ERROR
 ::= localValue 72

maxNumberOfMPTY-ParticipantsExceeded ERROR
 ::= localValue 126

resourcesNotAvailable ERROR
 ::= localValue 127

END
```

History

Document history	
February 1995	First edition
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